Qiskit v2.x Practice Quiz

Questions

1. Which gate maps $|0\rangle$ to $|+\rangle$?

   A. X
   B. Z
   C. H
   D. CX

2. What does an X gate do to $|0\rangle$?

   A. Creates $|+\rangle$
   B. Maps it to $|1\rangle$
   C. Measures the qubit
   D. Adds a global phase only

3. Which of the following is used for optimization considering code below? Assume all prerequisties were imported.

```
qc = QuantumCircuit(10)
qc.h(0)
for i in range(9):
    qc.cx(i,i+1)
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(2,backend=backend)
qc_isa = pm.run(qc)
```

   A. TrivialLayout
   B. CXCancellation
   C. CommutativeCancellation
   D. KAK Decomposition

4. Given the code below, how would you retrieve probability distribution from sampler results? Assume all components have been imported correctly.

```
qbits = QuantumRegister(2,"qbits")
clbits = ClassicalRegister(2,"clbits")
qc = QuantumCircuit(qbits, clbits)
(q0, q1) = qbits
(c0, c1) = clbits
qc.h(0)
```

```
qc.cx(0,1)
qc.measure(qbits, clbits)
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(backend=backend, optimization_level=2)
qc_isa = pm.run(qc)
sampler = SamplerV2(mode=backend)
job =sampler.run([qc_isa])
result = job.result()
```

A. counts = result[0].data.meas.get_counts()

B. counts = result[0].data.clbits.get_counts()

C. counts = result[0].data.evs.get_counts()

D. counts = result[0].data.stds.get_counts()

5. Which of the following are transpiler stages? *(Select all that apply)*

   A. Layout

   B. Routing

   C. Decomposing

   D. Scheduling

6. What is the purpose of transpilation?

   A. To convert a circuit into backend-compatible instructions

   B. To measure all qubits automatically

   C. To create a statevector

   D. To define an observable

7. Which of these is not a valid Estimator PUB entry?

   A. a single parametarized quantum circuit

   B. a single quantum circuit that contains no parameters

   C. target precision

   D. number of shots

8. Which of the following passes are used in heuristic stage of transpilation layout stage? *(Select all that apply)*

   A. TrivialLayout

   B. DenseLayout

   C. VF2Layout

   D. SabreLayout

9. You can access your submitted jobs by providing QiskitRuntimeService().jobs() method which of the following? *(Select all that apply)*

   A. date jobs were created after

B. date jobs were created before
C. CRN
D. API key

10. Why may SWAP gates be inserted during transpilation?

    A. To reset qubits
    B. To satisfy hardware connectivity constraints
    C. To increase shot count
    D. To remove all noise

11. What does the Bloch sphere represent?

    A. Only classical bits
    B. Single-qubit pure states
    C. Only two-qubit entangled states
    D. Classical probability distributions only

12. Which of the following state vectors corresponds to the following OpenQASM 3 code?

```
OPENQASM 3;
include "stdgates.inc";

qubit[2] q;
bit[2] c;

h q[0];
cx q[1], q[0];
z q[1];
```

    A. $\frac{\sqrt{2}}{2}|00\rangle + \frac{\sqrt{2}}{2}|01\rangle$
    B. $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$
    C. $\frac{\sqrt{2}}{2}|00\rangle - \frac{\sqrt{2}}{2}|01\rangle$
    D. $\frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle$

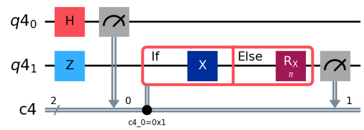
13. What does layout refer to in transpilation?

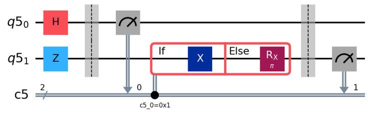
    A. Mapping virtual qubits to physical qubits
    B. Choosing the number of shots
    C. Drawing the circuit
    D. Choosing classical register names

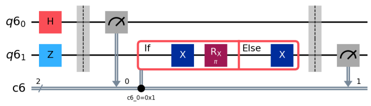
14. Which one of the following images is the output from the code below:

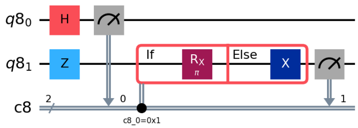
```
from qiskit import *
from math import pi

qubits = QuantumRegister(2)
clbits = ClassicalRegister(2)
```

```
qc = QuantumCircuit(qubits, clbits)
(q0, q1) = qubits
(c1, c2) = clbits

qc.h(0)
qc.z(1)
qc.barrier()
qc.measure(q0, c1)
with qc.if_test((c1,1)) as else_:
    qc.x(q1)
with else_:
    qc.rx(pi, q1)
qc.barrier()
qc.measure(q1,c2)
qc.draw(output="mpl", plot_barriers=False)
```

A. 

B. 

C. 

D. 

15. What is the result of applying Z to $|1\rangle$?

    A. $|0\rangle$
    B. $|1\rangle$
    C. $-|1\rangle$
    D. $|+\rangle$

16. Which of these are OpenQASM 3 features? *(Select all that apply)*

    A. while loops
    B. angle
    C. try/catch
    D. barrier

17. What are return values of the Estimator when the observable is Pauli Z on a single qubit?

    A. 0 or 1
    B. $|0\rangle$ or $|1\rangle$
    C. a real number between -1 and +1
    D. a real number between 0 and +1

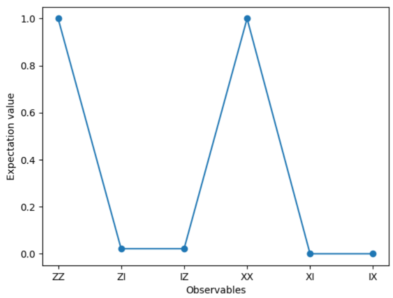
18. When is Sampler more appropriate than Estimator?

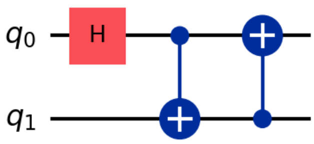
A. When you need bitstring measurement outcomes
B. When you need expectation values of observables
C. When you need to draw a circuit
D. When you need to create a backend

19. What does the Sampler primitive primarily return?

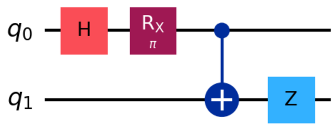
A. Expectation values
B. Measurement samples or bitstring results
C. Only statevectors
D. Only transpiled circuits

20. Which circuit is most consistent with the following Estimator output graph

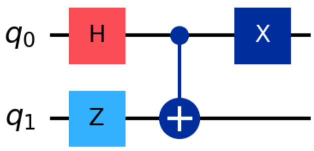




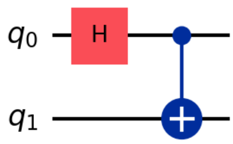
A.



B.



C.



D.

Answer Key

1. S1-002: C
2. S1-001: B
3. S3-005: C
4. S7-004: B
5. S3-007: A, B, D
6. S4-001: A
7. S6-005: D
8. S3-004: B, D
9. S7-007: A, B, C
10. S4-002: B
11. S2-001: B
12. S8-004: A
13. S4-003: A
14. S3-003: A
15. S1-003: C
16. S8-005: A, B, D
17. S6-003: C
18. S5-002: A
19. S5-001: B
20. S2-004: D